Eagle Delivery Project Report

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Problem Formulation

- State: row/col of a traversable cell
- Initial state: start of building's cell
- Actions: Up, Down, Left, Right
- Transition model: a valid move shifts one cell. Walls and out-of-bounds cells aren't legal moves in search, and the map is fully known and deterministic.
- Goal test: state equals the destination cell
- Path cost: the number of moves, with each costing 1

BFS is an optimal search algorithm because every step costs 1, so BFS expands nodes in order of increasing depth. The first time it reaches its goal, it has used the fewest possible moves.

Manhattan distance is also admissible because one move changes the row or column by at most 1, with no diagonals, so any route needs at least $|\Delta row| + |\Delta col|$ moves. Walls can only make the real route longer, never shorter, so h doesn't overestimate.

Map	Delivery Task	BFS Route Length	A* Route Length	BFS Expanded Node	A* Expanded Node
Baseline	Student Center -> Walton Hall	18	18	67	22
Baseline	Halle Library -> Downing Hall	22	22	79	62
Baseline	Pray-Harrold -> Putnam Hall	16	16	41	16
Detour	Student Center -> Walton Hall	26	26	70	51
Detour	Halle Library -> Downing Hall	22	22	65	66
Detour	Pray-Harrold -> Putnam Hall	16	41	16	16